THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 3: **Number Play** | Class 6 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 55	Reverse and add.
2	**(d)** 5175	Arrange digits in descending and ascending order and subtract.
3	**(c)** 9	Even: halve it. Odd: multiply by 3 and add 1.
4	**(d)** 2	Keep reversing and adding.
5	**(d)** 22	Add all the digits.
6	**(a)** 31	Add the two cells below.
7	**(d)** 19	Middle row: $a+b$ and $b+c$; top $= a+2b+c$.
8	**(d)** 77	Reverse and add.
9	**(c)** 7353	Arrange digits in descending and ascending order and subtract.
10	**(b)** 34	Even: halve it. Odd: multiply by 3 and add 1.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	47 each gets reversed and added (1); Palindrome reached: 121 after 1 step(s) (1).	1+1
12	Largest 9621, smallest 1269 (1); Difference $= 8352$ (1).	1+1
13	Even: halve it; odd: multiply by 3 and add 1 (1); $6, 3, 10, 5, 16, 8, 4, 2, 1$ (1).	1+1
14	58 then 143 each gets reversed and added (1); Palindrome reached: 484 after 2 step(s) (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Repeat the largest-minus-smallest step (1); $3524 \to 3087 \to 8352 \to 6174$ (1); Number of steps $= 3$ (1).	1+1+1
16	Follow the rule step by step (1); Sequence: 13, 40, 20, 10, 5, 16, 8, 4, 2, 1 (1); Steps $= 9$, maximum $= 40$ (1).	1+1+1
17	Digit sum $= 7 \times 5 = 35$ (1); Not divisible by 3, because the digit sum is not a multiple of 3 (2).	1+2
	OR: Repeat the largest-minus-smallest step (1); $4821 \to 7173 \to 6354 \to 3087 \to 8352 \to 6174$ (1); Number of steps $= 5$ (1).	1+1+1
18	Follow the rule step by step (1); Sequence: 14, 7, 22, 11, 34, 17, 52, 26, 13, 40, 20, 10, 5, 16, 8, 4, 2, 1 (1); Steps $= 17$, maximum $= 52$ (1).	1+1+1
	OR: Digit sum $= 8 \times 6 = 48$ (1); Divisible by 3, because the digit sum is a multiple of 3 (2).	1+2

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Step 1: 4311-1134=3177 (1); Step 2: result 6354 (1); Step 3: result 3087 (1); The numbers move toward the Kaprekar constant 6174 (2).	1+1+1+2
	OR: 121+232=353 (1); Palindrome (1); Start 29: 29, 121 (1); Palindrome 121 after 1 step(s) (2).	1+1+1+2
20	Step 1: 8721-1278=7443 (1); Step 2: result 3996 (1); Step 3: result 6264 (1); The numbers move toward the Kaprekar constant 6174 (2).	1+1+1+2
	OR: 343+454=797 (1); Palindrome (1); Start 38: 38, 121 (1); Palindrome 121 after 1 step(s) (2).	1+1+1+2

Mistakes to avoid:

- Writing 0 first when forming the smallest number (e.g. 046); the smallest 3-digit number from 4, 0, 6 is 406.
- In a Kaprekar step, subtracting the larger from the smaller or adding the two arrangements.
- Applying the odd rule ($3n+1$) to an even number or halving an odd number in the hailstone game.
- Reversing digits wrongly in reverse-and-add (the reverse of 68 is 86) or stopping one step early.
- Counting the starting number as a step when counting steps to reach 1 or a palindrome.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 3: **Number Play** | Class 6 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 3	Keep reversing and adding.
2	**(c)** 35	Add all the digits.
3	**(a)** 72	Add the two cells below.
4	**(a)** 16	Middle row: $a+b$ and $b+c$; top $= a+2b+c$.
5	**(a)** 99	Reverse and add.
6	**(b)** 6174	Arrange digits in descending and ascending order and subtract.
7	**(c)** 13	Even: halve it. Odd: multiply by 3 and add 1.
8	**(b)** 6	Keep reversing and adding.
9	**(a)** 22	Add all the digits.
10	**(a)** 94	Add the two cells below.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Largest 7430, smallest 347 (1); Difference $= 7083$ (1).	1+1
12	Even: halve it; odd: multiply by 3 and add 1 (1); $9, 28, 14, 7, 22, 11, 34, 17, 52, 26, 13, 40, 20, 10, 5, 16, 8, 4, 2, 1$ (1).	1+1
13	69 then 165 then 726 then 1353 each gets reversed and added (1); Palindrome reached: 4884 after 4 step(s) (1).	1+1
14	Largest 8531, smallest 1358 (1); Difference $= 7173$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Repeat the largest-minus-smallest step (1); $6352 \to 4176 \to 6174$ (1); Number of steps $= 2$ (1).	1+1+1
16	Follow the rule step by step (1); Sequence: 17, 52, 26, 13, 40, 20, 10, 5, 16, 8, 4, 2, 1 (1); Steps $= 12$, maximum $= 52$ (1).	1+1+1
17	Digit sum $= 4 \times 9 = 36$ (1); Divisible by 3, because the digit sum is a multiple of 3 (2).	1+2
	OR: Repeat the largest-minus-smallest step (1); $8730 \to 8352 \to 6174$ (1); Number of steps $= 2$ (1).	1+1+1
18	Follow the rule step by step (1); Sequence: 19, 58, 29, 88, 44, 22, 11, 34, 17, 52, 26, 13, 40, 20, 10, 5, 16, 8, 4, 2, 1 (1); Steps $= 20$, maximum $= 88$ (1).	1+1+1
	OR: Digit sum $= 5 \times 4 = 20$ (1); Not divisible by 3, because the digit sum is not a multiple of 3 (2).	1+2

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Step 1: 6210-126=6084 (1); Step 2: result 8172 (1); Step 3: result 7443 (1); The numbers move toward the Kaprekar constant 6174 (2).	1+1+1+2
	OR: 565+676=1241 (1); Not a palindrome (1); Start 49: 49, 143, 484 (1); Palindrome 484 after 2 step(s) (2).	1+1+1+2
20	Step 1: 8742-2478=6264 (1); Step 2: result 4176 (1); Step 3: result 6174 (1); The numbers move toward the Kaprekar constant 6174 (2).	1+1+1+2
	OR: 787+898=1685 (1); Not a palindrome (1); Start 56: 56, 121 (1); Palindrome 121 after 1 step(s) (2).	1+1+1+2

Mistakes to avoid:

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- In a Kaprekar step, subtracting the larger from the smaller or adding the two arrangements.
- Applying the odd rule ($3n+1$) to an even number or halving an odd number in the hailstone game.
- Reversing digits wrongly in reverse-and-add (the reverse of 68 is 86) or stopping one step early.
- Counting the starting number as a step when counting steps to reach 1 or a palindrome.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 3: **Number Play** | Class 6 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 18	Middle row: $a+b$ and $b+c$; top $= a+2b+c$.
2	**(c)** 132	Reverse and add.
3	**(c)** 7083	Arrange digits in descending and ascending order and subtract.
4	**(a)** 22	Even: halve it. Odd: multiply by 3 and add 1.
5	**(b)** 4	Keep reversing and adding.
6	**(d)** 15	Add all the digits.
7	**(d)** 111	Add the two cells below.
8	**(d)** 24	Middle row: $a+b$ and $b+c$; top $= a+2b+c$.
9	**(c)** 154	Reverse and add.
10	**(d)** 46	Even: halve it. Odd: multiply by 3 and add 1.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Even: halve it; odd: multiply by 3 and add 1 (1); $10, 5, 16, 8, 4, 2, 1$ (1).	1+1
12	78 then 165 then 726 then 1353 each gets reversed and added (1); Palindrome reached: 4884 after 4 step(s) (1).	1+1
13	Largest 7652, smallest 2567 (1); Difference $= 5085$ (1).	1+1
14	Even: halve it; odd: multiply by 3 and add 1 (1); $12, 6, 3, 10, 5, 16, 8, 4, 2, 1$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Repeat the largest-minus-smallest step (1); $2194 \to 8172 \to 7443 \to 3996 \to 6264 \to 4176 \to 6174$ (1); Number of steps $= 6$ (1).	1+1+1
16	Follow the rule step by step (1); Sequence: 20, 10, 5, 16, 8, 4, 2, 1 (1); Steps $= 7$, maximum $= 20$ (1).	1+1+1
17	Digit sum $= 2 \times 12 = 24$ (1); Divisible by 3, because the digit sum is a multiple of 3 (2).	1+2
	OR: Repeat the largest-minus-smallest step (1); $5867 \to 3087 \to 8352 \to 6174$ (1); Number of steps $= 3$ (1).	1+1+1
18	Follow the rule step by step (1); Sequence: 22, 11, 34, 17, 52, 26, 13, 40, 20, 10, 5, 16, 8, 4, 2, 1 (1); Steps $= 15$, maximum $= 52$ (1).	1+1+1
	OR: Digit sum $= 9 \times 3 = 27$ (1); Divisible by 3, because the digit sum is a multiple of 3 (2).	1+2

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Step 1: 9530-359=9171 (1); Step 2: result 8532 (1); Step 3: result 6174 (1); The numbers move toward the Kaprekar constant 6174 (2).	1+1+1+2
	OR: 161+272=433 (1); Not a palindrome (1); Start 67: 67, 143, 484 (1); Palindrome 484 after 2 step(s) (2).	1+1+1+2
20	Step 1: 5544-4455=1089 (1); Step 2: result 9621 (1); Step 3: result 8352 (1); The numbers move toward the Kaprekar constant 6174 (2).	1+1+1+2
	OR: 383+494=877 (1); Not a palindrome (1); Start 79: 79, 176, 847, 1595, 7546, 14003, 44044 (1); Palindrome 44044 after 6 step(s) (2).	1+1+1+2

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